

Leak Protocol

A permissionless platform to tokenize everything.

DIGITAL RWA AVALANCHE C-CHAIN UNISWAP V4

01

Overview

Any digital asset with a provable owner becomes a token with a live market, in one transaction, for about a cent.

Digital RWA is the asset class: real assets, real owners, real demand signals, no legal wrapper. Media and social interaction came first, GitHub made code an addressable unit, and **AI assets** — agents, skills, models, datasets — changed the shape of the category. Each is independently owned, independently forked, independently measured. None has a price.

Leak Protocol is two layers, and the front tier is the one that matters to a user. The **asset tokenization layer** classifies a digital asset, proves who owns it through the platform that already custodies it, and mints against that proof. It also picks what the market settles in.

Beneath it, a **token launch platform** mints supply straight into a single Uniswap v4 pool – tradable in the next block, no bonding-curve phase, no graduation threshold, no migration. That layer is permissionless: any platform can tokenize through it without an agreement and without depending on a server Leak Protocol operates. Curve shape and pair currency are both parameters, so each asset class gets a market built for it, and every trade splits across five roles on-chain.

\$0.01

COST TO OPEN A MARKET

0s

WAIT BEFORE THE FIRST
TRADE

5

ROLES PAID FROM ONE 1%
FEE

0.198%

OF GRADUATION-MODEL
LAUNCHES EVER REACH A
MARKET

AI assets are produced faster than they can be funded

Agents, skills, models and datasets became individually addressable products in under two years, and each has a venue: agents on [SeekClaw](#), skills on [Skills](#), models and datasets on [HuggingFace](#) – forked, fine-tuned, benchmarked and re-published in the open by people who are not employed to do it and are not paid when it works.

[Virtuals Protocol](#) settled the demand side: people will buy a token representing an AI agent. The gap is on the supply side – the builders producing these assets have no instrument that converts adoption into capital at their size.

Graduation is failing, measurably

Every mainstream launchpad holds a token on a private curve until reserves cross a fixed threshold, then migrates it to a real AMM.

SAMPLE	LAUNCHES	GRADUATED	RATE
Sept–Oct 2025	655,770	~4,130	0.63%
8 May – 10 Jun 2026	832,941	~1,650	0.198%

[PumpDotFun](#) launches · 2026 figure Wilson 95% CI 0.189–0.208% · a 3.18× decline in eight months

WHAT THE NUMBER MEANS

~831,300 assets paid the launch cost and died inside a curve: priced by a formula, with liquidity that could not leave. The threshold is not a quality filter. It is a minimum viable size, and it sits far above where most assets live.

It persists because it is load-bearing. Graduation funds the migration: the curve accumulates reserves, the reserves seed the pool, and the threshold is where reserves finally cover the flat cost of moving them. Lower it and that cost is paid on tokens whose lifetime fees never repay it. Remove it and the curve phase has no purpose — which deletes the pricing engine, the reserve custody and the graduation event the model is built around.

The design's grip is visible in its successors. [Bags](#) rebuilt the creator economics on top of it — mandatory, perpetual fee sharing configured at launch — and kept the curve and the migration intact, because removing them removes the product.

Leak Protocol has no threshold because it has no migration. Supply is minted directly into the pool as single-sided liquidity, so no counter-currency is needed to open a market and the entire cost is one transaction.

RWA arrived, but only at one end

Tokenized real-world assets crossed \$3B on Avalanche — securities, credit and funds, behind institutional compliance. The long tail of digital assets received none of it, because every step of that pipeline assumes a legal wrapper.

Digital RWA

The segment has an observable boundary: **the asset has a verifiable owner and no legal wrapper.**

CLASS	PLATFORM	OWNERSHIP PROVEN BY	PUBLIC SIGNAL
Agent	SeekClaw	platform auth	usage, installs
Skill	Skills	platform auth	installs
Model, dataset	HuggingFace	platform auth	downloads, forks
Repository	GitHub	platform auth	stars, forks
Social post	X	platform auth	reach
Short media	Leak Platform IPFS	uploader attestation	views
NFT	any chain	the chain itself	floor, holders

These are real assets with real off-chain existence. What no token minted against them does is convey a legal claim – the token is a market on the asset's attention, not a share

in it.

WHY THE BOUNDARY IS THE DESIGN

Because no Digital RWA carries a legal wrapper, none crosses the perimeter that makes institutional tokenization slow, expensive and permissioned. The constraint and the capability are the same fact: that is what allows this layer to be permissionless.

What tokenizing gives the owner

A maintainer with 400 stars, an agent developer with 2,000 users, a model author whose fine-tune gets forked weekly: none can convert adoption into capital at that size. Grants need an application, sponsorship needs an audience, equity needs a company.

Tokenizing does it in one transaction — **1% of supply at mint, then 0.50% of every dollar ever traded, permanently.** The stream scales with exactly the thing these builders already produce.

What the buyer is buying

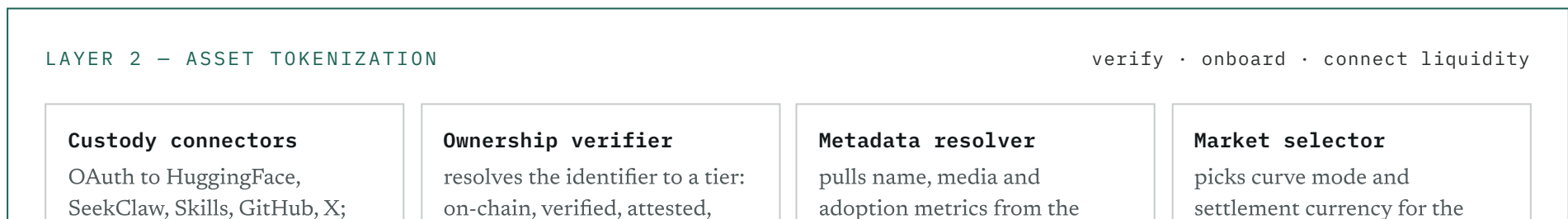
The negative first, because it makes the rest credible: not equity, not a revenue share, not a licence, no governance. It is a position on the asset's attention.

PROPERTY	WHY IT HOLDS
Something to form a view about	stars, downloads, installs and reach are published by the custodian and produced by parties with no stake in the token – an observable series, not a narrative
Owner incentive points the same way	0.50% of volume for the life of the market and 1% of supply from mint, with no vesting cliff to sell into
The exit does not decay	20% of every fee is re-minted as liquidity nobody can withdraw, so day-one depth is a floor rather than a peak

04

Architecture

Two layers over the Uniswap v4 singleton, and the stack is read top-down. **Layer 2 is the front tier** – where a user arrives and where an asset becomes a token. Layer 1 is the settlement tier beneath it: a public primitive that knows nothing about what a token represents.



IPFS for direct uploads

reject

custodian API

asset class

```
deploy(owner, proof, uri, curveMode, currency, salt)
```



LAYER 1 – TOKEN LAUNCH PLATFORM · PERMISSIONLESS

one pool per asset

Factory

CREATE2 deploy,
predicts the coin
address before signing

Coin – ERC-20

mints 1B: 1% to the
owner, 99% to the hook

Curve registry

poolConfig bytes per
mode; pays the mode's
author 0.01%

Currency registry

admitted settlement
assets; enforces one
hop from the base asset

Hook

afterInitialize mints the
ladder · beforeSwap
sets the dynamic fee ·
afterSwap splits it

Router

native-AVAX entry,
passes the trade referrer
through to the hook

```
initialize(poolKey, sqrtPrice) · mint 31 single-sided positions
```



UNISWAP V4 POOLMANAGER

singleton · settlement

Pool

one per asset, dynamic-fee flag,
tick spacing 200

Ladder + tail

nested positions to the
discovery top, one tail to
MAX_TICK

Fee accrual

collected on every swap, routed
back to the hook

Quoter

the only price source;
simulated per quote

Layer 1 is a public primitive. Every call is built client-side by the SDK: no API key, no rate limit, no server that can change terms. Any platform can call `deploy` directly.

Layer 2 is the reference implementation of the caller. It holds no privilege inside layer 1 — it earns the same integrator fee leg as any third party that builds the same thing.

THE SEPARATION IS THE POINT

A launch platform only its own front end can use is an app. One that pays outsiders to build on it, automatically, is a layer.

Layer 2 — Asset tokenization

The front tier. Every user arrives here, every asset becomes a token here, and every market gets the currency it settles in here. The launch platform underneath (§6) is indifferent to what a token represents — this is the layer that decides whether any of it is a market or a casino.

JOB	WHAT IT DOES	WHY IT DECIDES THE OUTCOME
Verify	reads ownership from the platform that already custodies the asset	a token bound to a proven owner is a different instrument from a token bound to a name anyone can type
Onboard	the owner signs in to a platform they already use – no new account, no listing application, no review queue	the acquisition surface is the custodian's existing user base, not ours
Connect liquidity	selects the settlement currency and curve mode at mint	the pair sets the ceiling on depth and decides who can trade the asset without an extra hop

Verification: three tiers, assigned once

TIER	CUSTODY SOURCE	PROOF	STRENGTH
On-chain	any chain	the connected wallet holds the token	cryptographic
Verified	HuggingFace · SeekClaw · Skills · GitHub · X	the custodian's own auth reports the owner	as strong as that platform
Attested	Leak Platform IPFS	the uploader signs the claim	a claim, labelled as one
Reject	—	ownership cannot be established	not minted

The tier is written on-chain and shown wherever the asset appears. Attested is never described as verification – that distinction is the product, and collapsing it would make every other claim in this document worthless.

Liquidity: the settlement currency is a Layer 2 decision

Layer 1 exposes the currency as a parameter (§6). Layer 2 chooses it, per asset class, and that choice is what connects a coin to capital that already exists rather than capital that has to be recruited.

SETTLEMENT CLASS	EXAMPLES	STATUS	WHAT IT UNLOCKS
Native	AVAX	default	one hop from the chain's base asset; depends on no other market
Flagship crypto	BTC.b · HYPE	phase 2	the deepest non-native books, and holders who never had to acquire AVAX first
Tokenized equity	NVDA · SPCX	phase 2	denominates the coin in a sector – the coin stops being an absolute bet and becomes a relative one

The registry admits any ERC-20 that clears a depth floor, under the Layer 1 rules (§6): one hop from the base asset, never the protocol's own token as default

WHY TOKENIZED EQUITY IS THE INTERESTING ONE

An open model quoted in AVAX asks whether people like the model. The same model quoted in NVDA asks whether open weights outperform the chip incumbent – a question with a thesis behind it, an existing audience, and a counterparty who already holds the other side. An agent quoted against a software-sector asset asks the same about a category. That instrument does not exist anywhere today, and nothing about it requires new protocol code: it is a currency-registry entry.

Trust assumptions

ASSUMPTION	CONSEQUENCE IF IT BREAKS
Custodian auth reports the true owner	tokens minted through that platform are compromised – blast radius is that platform alone, and the tier is on-chain per asset
Ownership at mint is what counts	later transfer, suspension or deletion of the account does not retroactively change the token
No tier conveys a legal claim	tokenizing a repository or a model does not transfer, license or encumber it
Quotes come only from the on-chain quoter	a pool whose hook affects swaps cannot be priced locally, so integrators need an RPC call per quote

Layer 1 — Token launch platform

There is no bonding curve

99% of supply is minted straight into one Uniswap v4 pool as single-sided liquidity, held by a hook. Price moves because the AMM crosses ticks – no pricing formula, no reserve contract. The pool opens at the bottom of its range, so every position is

reserve contract. The pool opens at the bottom of its range, so every position is denominated purely in the coin.

Positions are *nested* rather than adjacent: each carries an equal share of coin over a strictly shorter range ending at a common far tick. Depth is therefore non-decreasing in price, and the currency needed to move price by a fixed proportion strictly increases as the market grows.

ACTOR	ENTRY	EXIT	CURRENCY IN	COINS OUT	COST PER +10%
Alice	1.311e-4	2.119e-4	964.67	5,430,624	192
Bob	2.119e-4	3.493e-4	4,431.88	15,656,279	845
Carol	9.122e-4	1.070e-3	207,358.40	207,386,758	123,866

Reference multicurve configuration, reproduced with tools/curve/ · currency units generic

Moving price 10% costs Carol **647× what it cost Alice**. Early entry is rewarded with price, late entry with depth: the book becomes hard to move exactly as it becomes worth moving.

Above the last discovery tick a single tail position runs to the maximum tick. That tail replaces graduation – the pool cannot run out of coin at any price, and no position is withdrawable, so liquidity cannot be pulled.

Both sides of the book

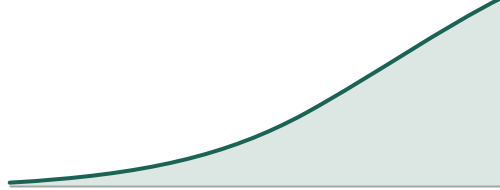
Selling walks the same ladder in reverse. Liquidity at a given tick is identical in both directions, so a seller near the top meets the deepest part of the book and impact per coin grows as price falls back toward the opening tick. There is no redemption path and no reserve to drain: every exit is a swap against the same pool every entry went through.

SOLVENCY INVARIANT

Currency payable out can never exceed currency paid in, less fees already distributed. The pool holds no leverage, borrows nothing, and promises nothing it is not already holding. There is no state in which a holder is owed something the pool does not have — which is the whole reason the protocol can be permissionless about what gets listed.

Curve modes are data, and their authors are paid

The curve is a `poolConfig` byte string — version, currency, and per-curve arrays of tick bounds, position counts and supply shares. A new mode is a new configuration: no contract, no upgrade, no governance vote. That makes curve design a role rather than an internal decision, and the protocol pays it: **whoever authors the mode a coin launches with earns 0.01% of that coin's notional volume for the life of the market.**

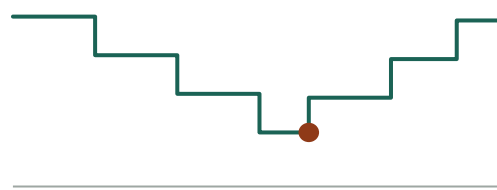


Multicurve

DEFAULT · LIVE

Several curve segments blended into one book. Deep near spot, steep in the tail.

KNOWN REFERENCE PRICE — MEDIA,
POSTS, REPOSITORIES

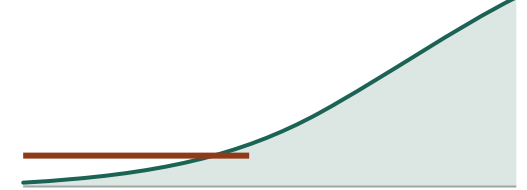


Dynamic Dutch

PLANNED

Price decays each epoch until demand meets it, then ratchets up.

NO OBVIOUS OPENING PRICE — NEW
MODELS, FIRST-RUN AGENTS

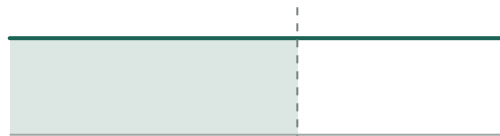


Lockable multicurve

LIVE

Multicurve plus a permanently locked LP position. Signals that liquidity cannot be pulled.

SOLD ON TRUST — DATASETS, SKILLS,
LONG SUPPORT HORIZONS



Fixed price

PLANNED

One price until the depth target fills, then a flat book. Reads like a presale, no discovery.

EXTERNALLY SET VALUE — NFTS,
LICENSED MODELS

Vertical axis: price. Horizontal: cumulative supply sold.

Shapes are schematic.

The pair currency is the second parameter

A pool opens single-sided: 100% coin, zero currency. Every unit of the currency side has to be bought in. So the asset a coin settles against decides who can trade it without an extra hop, what unit its price is quoted in, and what the creator is actually paid in. Which makes the pair a ceiling: **a coin can never be deeper than the asset it settles against.**

ONE HOP	asset coin	AVAX		
CHAINED PAIRS	asset coin	another coin	protocol token	base asset
RULE	BECAUSE			
One hop from the base asset	a coin paired against another launched coin inherits that coin's depth as its ceiling; the weakest link caps everything downstream and routers cannot reliably find the path			
Never the protocol's own token as default	it forces two exposures at once – the asset's move and the token's beta – so a coin can rise against the token and still fall in dollars			
Denomination is a product decision	against a stable unit the coin is an absolute bet; against a sector asset it is a relative one			

A Digital RWA quoted against the asset class it competes with. An open model priced against a chip-sector asset asks whether the open model outperforms the incumbent. That is a legible thesis rather than a punt, and it falls straight out of currency being a parameter instead of a constant.

Phase 1 settles everything in native AVAX. The field is in the pool config from the first block, so opening it later is a registry change, not a migration.

The launch fee pays the creator

The pool runs a dynamic fee: 99% at creation, decaying linearly to 1% over 30 seconds. It is a *pool* fee, so none of it is burned — it flows into the same distribution as any other trade.

T	0S	3S	15S	≥30S
fee	99.00%	89.20%	50.00%	1.00%
DANA BUYS AT T = 3S WITH 100 AVAX • FEE 89.20%				AVAX
Re-minted as permanent liquidity				17.84
Creator				44.60

Platform that created the coin	17.84
Platform owner	4.46
Trading referrer	3.57
Curve owner	0.89
Dana's actual purchase	10.80

Sniping inside the launch window is a transfer from the sniper to the creator and to permanent liquidity. There is no anti-bot heuristic in the protocol – bots are made to pay for the launch. The creator's own first buy is exempt, which is why deploy and first buy are a single transaction.

Five roles, one fee

LIQUIDITY 20%	CREATOR 50%			PLATFORM 20%				
LIQUIDITY 20%	CREATOR 50%	PLATFORM 20%	OWNER 5%	REFERRER 4%	CURVE 1%			
ROLE	NOTIONAL	PER \$1M VOLUME	PAID FOR					
Permanent liquidity	0.20%	\$2,000	re-minted, unwithdrawable – depth grows with volume					
Creator	0.50%	\$5,000	owning the asset					

Third-party platform	0.20%	\$2,000	bringing the asset
Platform owner	0.05%	\$500	running the protocol
Trading referrer	0.04%	\$400	bringing the trade
Curve owner	0.01%	\$100	designing the market's shape

Multi-recipient fee routing is proven, not speculative: Bags has settled over **\$5B** of cumulative volume with fee sharing made mandatory at launch, paid to designated recipients in perpetuity. The difference here is *what the recipients are*. Bags splits to wallets a creator names. Leak Protocol splits to **roles defined by a function performed** – and one of them has no analogue anywhere: no other launchpad pays for the shape of the market, because no other launchpad lets the shape vary.

The re-minted 20% never leaves. An asset's liquidity becomes a function of volume already done rather than of anyone's continued willingness to provide it. **The third-party leg is the business model:** a platform tokenizing its assets through Leak Protocol earns 0.20% of notional on everything it brings, with no agreement to sign and no revenue share to invoice.

An asset doing \$50,000 of lifetime volume returns \$250 to its owner, \$100 to that platform, \$25 to the protocol, \$5 to the curve author, and adds \$100 of liquidity that can never be removed – against a \$0.01 launch cost.

Related work

PROTOCOL	UNIT TOKENIZED	OWNERSHIP PROVEN	TRADABLE AT LISTING	CURVE SHAPE	FEE RECIPIENTS
Pump.fun	a name	no	no – 85 SOL threshold	fixed	creator, 0.05%
Bags	a name + social	no	no – DBC migration	fixed (Meteora DBC)	creator + named wallets, 1%
Virtuals Protocol	an agent	no	no – 42,000 VIRTUAL	fixed	protocol
Clanker	a string	no	yes	fixed	protocol, 20% of LP fees
Zora	a post in Zora	Zora profile	yes	closed endpoint	creator
The Arena	a social account	Arena account	no – curve first	fixed	creator
Leak Protocol	a custodied asset	yes, by its custodian	yes	a parameter	five protocol roles, incl. the curve author

Pump.fun created this category and still defines it. Roughly **75–80% of all Solana launches** run through it, and the bonding-curve-then-migrate design that every launchpad here inherits – including the ones that criticise it – is its contribution. The argument in §2 is aimed at the threshold, not at the model: without Pump.fun there is no reference point for what a permissionless launch looks like at all. It revised its creator fee schedule in January 2026 after a volume spike exposed how thin the original incentive was, at 0.05% of trading fees.

Bags is the closest existing work on fee routing, and the reason a five-way split is an extension rather than a leap. Live since May 2025 on Meteora's dynamic bonding curve, it makes fee-sharing configuration *mandatory at launch*: 1% of trading volume flows to designated recipients in perpetuity, and a creator can name additional claimers. Over **\$5B** in cumulative on-chain volume. What it does not do is bind the token to anything but a social account, remove the curve and the migration, or pay anyone for designing the market – the curve is Meteora's, and it is the same curve for every token launched.

Virtuals Protocol proved demand for agent tokens but tokenizes agents only, and its curve is a constant. **Clanker** dropped the threshold and the ceremony – deploy an ERC-20 from a sentence, take 20% of pool-level LP fees, over \$8B cumulative volume – but makes no claim about what the token represents. A deployer, not a tokenization layer.

Zora is the closest existing mechanism and the honest baseline: a Uniswap v4 hook, a single-sided multi-curve ladder, a 1% pool fee, a dynamic launch fee. Threshold-free

launching is not claimed here as an invention – Zora shipped it. What Zora does not do is bind a coin to an asset held by a platform it does not own, expose curve shape as a caller-supplied parameter, or let an integrator build without routing through its infrastructure.

The Arena is the evidence the audience exists on this chain: 200,000+ registered users, \$450M+ volume and 35,000+ AVAX in fees within two months of relaunch, from a curve launchpad inside a social feed. A candidate integrator, not a competitor.

DELTA, STATED NARROWLY

Ownership proven against third-party custodians · curve shape as a per-class parameter whose author is paid · fee routing extended from creator-named wallets to **protocol roles, one of which is the market designer** · no protocol-operated server in the integration path.

Parameters

PARAMETER	VALUE	NOTE
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Total supply	1,000,000,000	fixed at mint, no further issuance
To market as liquidity	99%	held by the hook, unwithdrawable
To creator at mint	1%	transferable immediately
Pool fee	1.00%	dynamic flag, hook-controlled
Launch fee	99% → 1%	linear decay over 30s
Tick spacing	200	
Discovery positions per curve	11	reference config: 3 curves
LP positions minted	34 → 31	after overlap dedupe
Backing currency	AVAX	native; parameterised for future pairs
Deploy cost	~7M gas	0.00035 AVAX at 0.05 gwei

Two invariants are enforced on-chain at deploy: the sum of discovery supply shares must be strictly below 100% of supply, and total positions must fall between 2 and 200. Violations revert.

Roadmap

Phase 1 ships the primitive and proves it by being its own first caller. Phase 2 widens what can be tokenized and what it can be priced in. Nothing in phase 2 is required for the protocol to function.

PHASE 1 Both layers live on mainnet

LAYER 1	Permissionless launch platform on Avalanche mainnet. Multicurve and lockable multicurve live, five-way fee split settling on-chain, curve and currency registries deployed. Anyone can call <code>deploy</code> without asking.
LAYER 2	Custodian-verified tokenization for Agents, Skills, GitHub repositories, X posts and Short Media – ownership proven at mint, tier written on-chain.
SDK	Both layers published as open interfaces. A third party can launch a market through Layer 1 and verify ownership through Layer 2 from their own front end, building every call client-side – no API key, no allowlist, and no server operated by Leak Protocol anywhere in the path.
ASSURANCE	Third-party audit published; contracts verified on the explorer.

PHASE 2 Wider assets, wider money

LAYER 1	Dynamic Dutch and Fixed price modes. Curve authoring opened as a public role – anyone can publish a mode and earn 0.01% of the notional volume of every coin launched with it.
LAYER 2	HuggingFace models and datasets; NFT .
CURRENCY	Settlement opened well beyond AVAX: USDC for a stable unit, BTC.b and HYPE for the deepest crypto books, and tokenized equities – NVDA , SPGY – as an asset can be created against the master it represents with

tokenized equities – NVDA, SPCX – so an asset can be quoted against the sector it competes with.

GOVERNANCE

LEAK admitted as an *optional* pair currency. Never the default, for the reason given in §6.
